What mechanisms could explain masking?

Research documents ability concealment among gifted learners.

Patterns vary across age and social settings.





We examine possible learning mechanisms.

Swiatek (1995) · Chan (2005) · Foust, Rudasill & Callahan (2006)

A close second

“Second. Close second, yeah!”

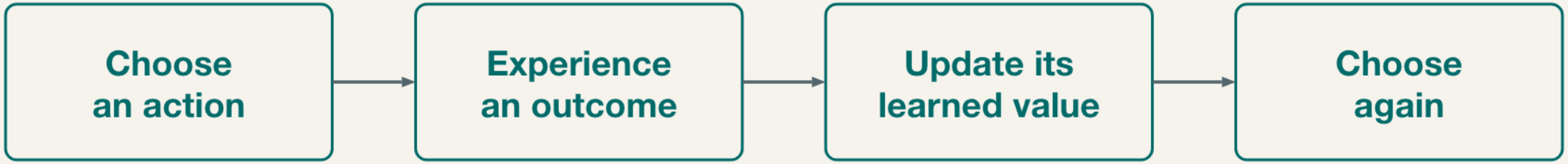


Social expectations can shape displayed ability.

The Incredibles (2004) · Pixar / Walt Disney Pictures · Film illustration

Learning from social experience

Experience changes expectations, which influence the next choice.



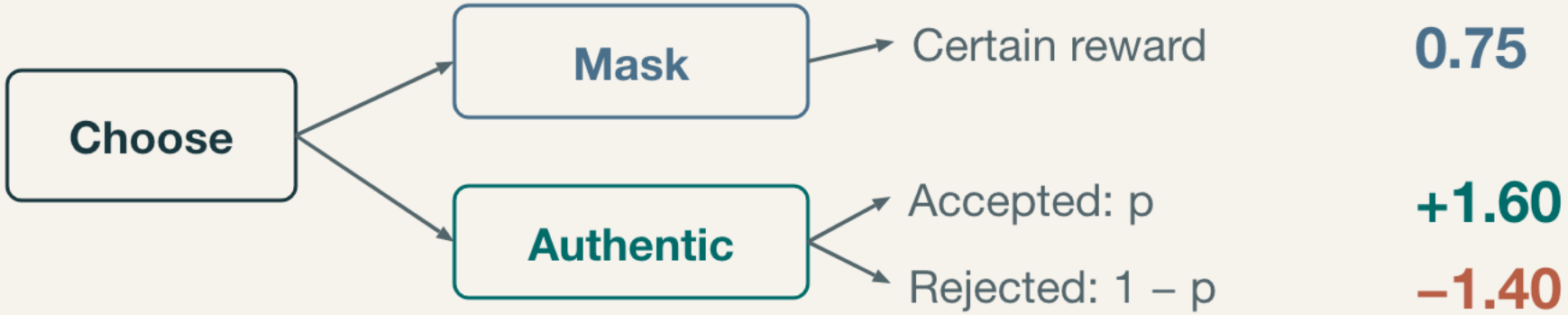
Baseline feedback concerns the chosen action.

Higher-valued actions become more likely.
Exploration continues.

A step is one interaction, not a calibrated year. Negative outcomes can receive extra weight.

M0 · Masking or authenticity

Two actions • fixed receptivity p • no persistent social state



THEORY

Higher expected payoff for authenticity: p > .717

At p=.75: 0.85 versus 0.75

SIMULATIONS

–2.58 pp authentic sampling

After initial rejection · 95% CI [–3.58, –1.59]

–11.81 pp final masking

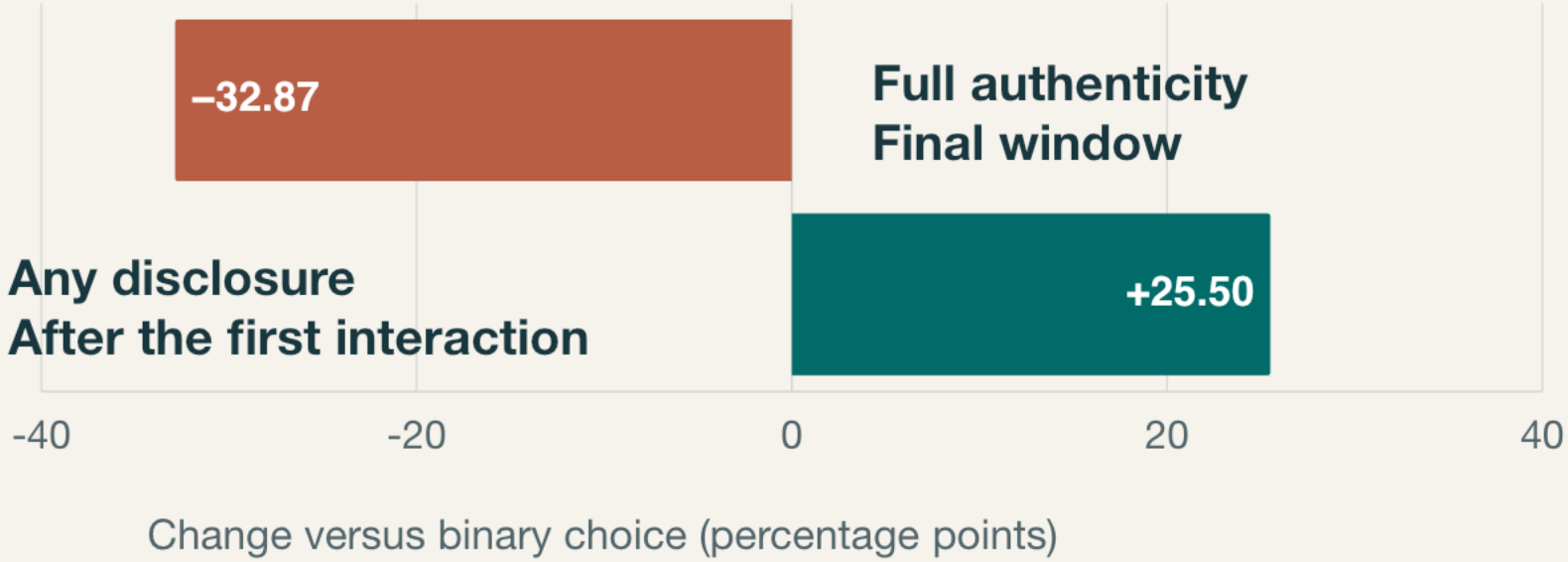
Extra feedback · 95% CI [–12.66, –10.97]

E01: p=.75/.85 pooled, λ=1; 500 learners/cell/master, T=2000, five masters. History final-window CI spans zero.

M1 · Participation and full expression differ

Five independently learned disclosure levels; no transfer between action values.

Disclosure 0 ¼ ½ ¾ 1



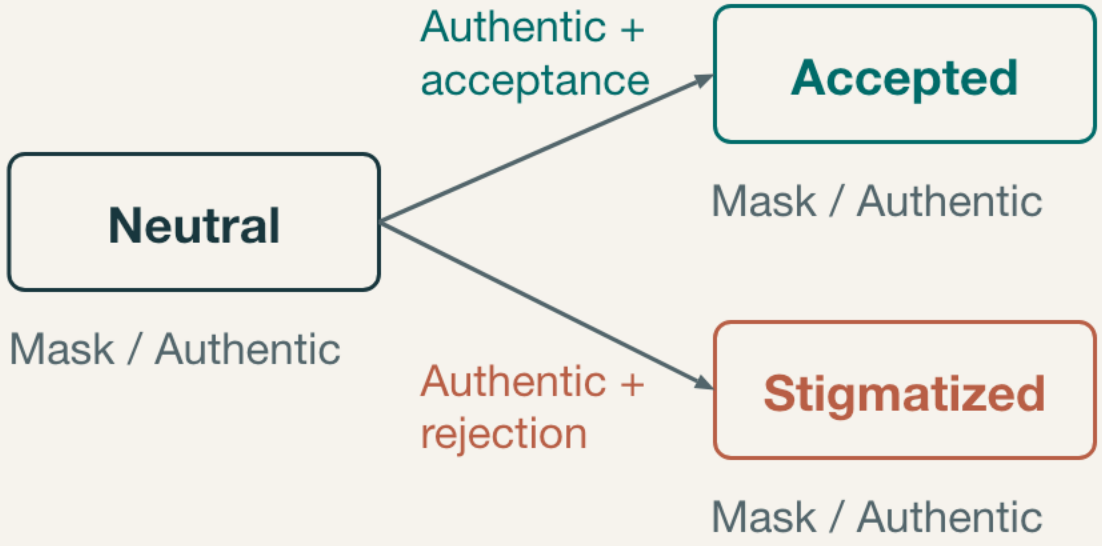
More participation can coexist with less full expression.

Theory: the payoff is linear in disclosure; the full endpoint is best in these cells.

E02: after initial rejection; p=.75/.85 pooled; five masters. CIs: [25.30,25.71] and [–34.20,–31.53] pp.

M2 · Persistent social consequences

Actions: Mask / Authentic in every state • Learner: 3 × 2 state–action values



THEORY · Neutral authenticity threshold

Static M0	M2: η=0	M2: η=.8
.717	.384	.269

Turnover returns Accepted to Neutral.
Escape can return Stigmatized to Neutral;
redemption can establish acceptance.

243 / 357

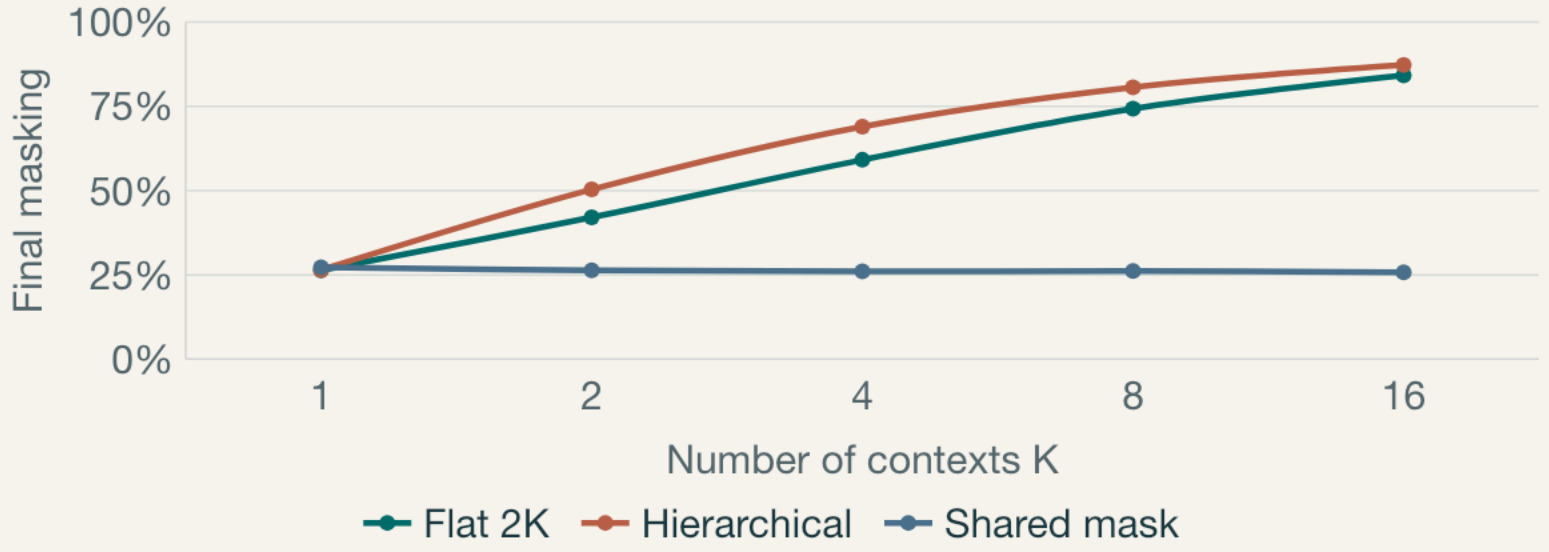
settings favour learned masking
while the dynamic optimum
favours authenticity

243/357 counts parameter settings, not learners. Neutral-state comparison; full-information discounted benchmark.

M3 · Context is part of the choice

Flat model: choose one of 2K group–behaviour pairs • group receptivities stay fixed

Group 1	Group 2	… Group K
$Mask_1$ / $Authentic_1$	$Mask_2$ / $Authentic_2$	$Mask_k$ / $Authentic_k$



At K = 16, flat model

84.19%

final masking

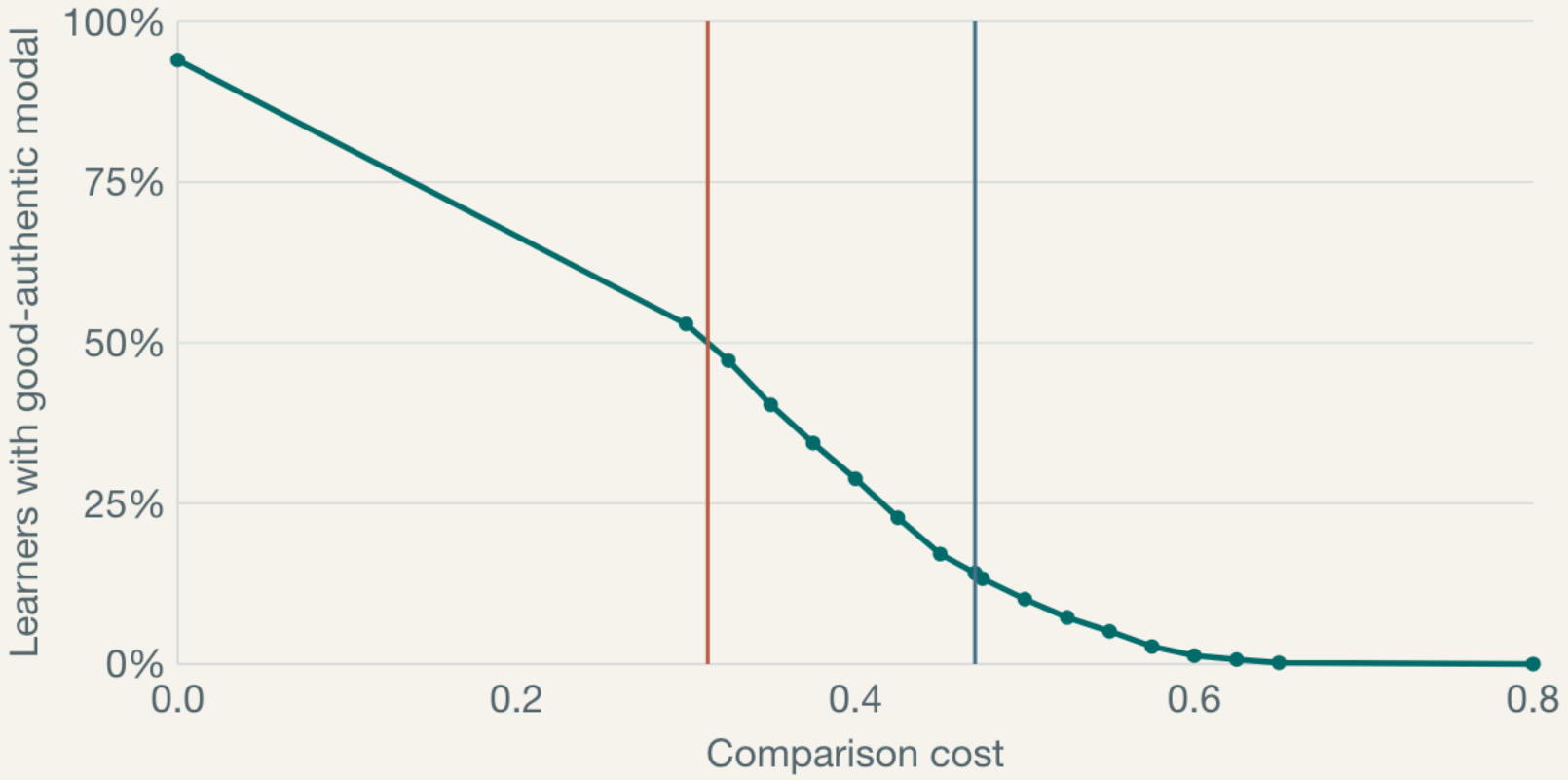
99.72%

had experienced acceptance in the good context

E04: one good context, fixed identity; five-master means. Encounter is over the full run; masking is the final-window action fraction.

M3 · Social comparison changes the boundary

Comparison cost reduces accepted authenticity in the receptive group.



LEARNED

0.313

95% CI [.308, .317]

OBJECTIVE

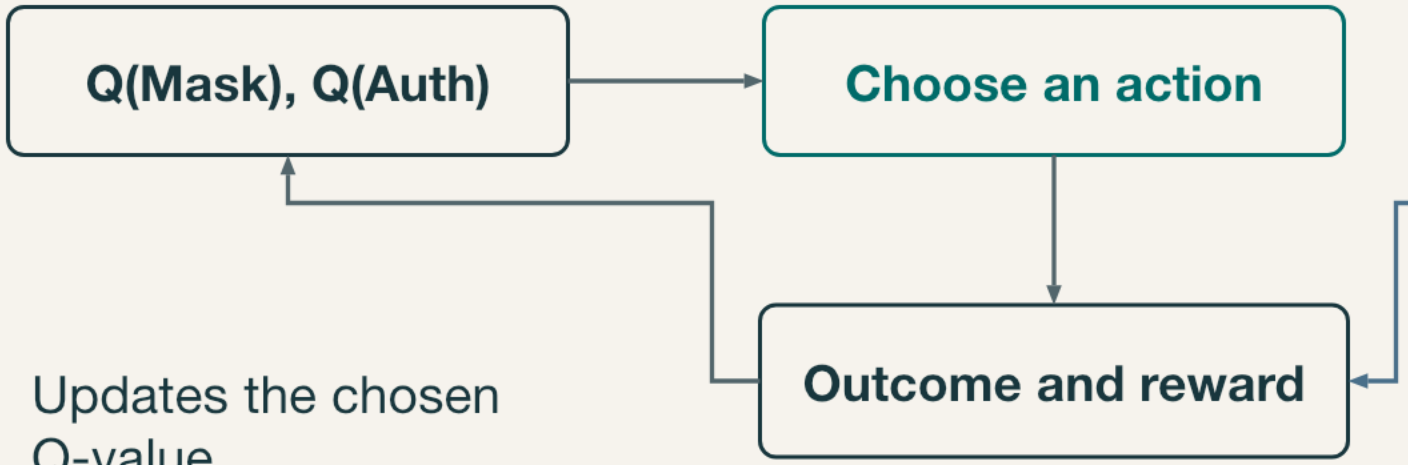
0.471

Expected payoffs are equal

E05: flat K=4, λ=1, five masters. Learned crossing is a modal-population boundary, not majority-masking onset.

M4 · Behaviour changes future receptivity

Actions: Mask / Authentic • Environment: continuous p_t • Learner: only two Q-values



Updates the chosen Q-value

Event	Next receptivity
Accepted authenticity	Increases
Rejected authenticity	Decreases
Masking	Decreases

p_t is bounded between .05 and .95.
The learner does not condition Q on p_t.

Implemented extension; formal E06 results deferred. Direction-of-change rules are model assumptions; hysteresis is not established.

What changes across the models?

Action = a choice • State = the current environmental condition

Model	Actions	Environmental state	Learner memory
M0	2 behaviours	Fixed receptivity	2 action values
M1	5 disclosure levels	Fixed receptivity	5 action values
M2	2 behaviours	3 social states	3 × 2 values
M3, flat	2K group–behaviour pairs	Fixed group receptivities	2K action values
M4	2 behaviours	Continuous receptivity p_t	2 action values

E01–E05: five independent master replications. Findings concern specified models; mechanisms in human learners remain empirical questions.

Different mechanisms imply different questions

What does the child expect?

Which experiences change that expectation?

Does learning transfer between situations?

What consequences persist?